

Section A : Pure Mathematics [40 marks]

- 1** The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -1 \\ 3 \end{pmatrix}$, where λ is a parameter. The point A has coordinates $(5, 1, 3)$.
- (a) Show that the point A does not lie on the line l_1 . [1]
- (b) The plane Π_1 contains the point A and the line l_1 . Find the Cartesian equation of the plane Π_1 . [3]
- (c) The plane Π_2 is obtained from the plane Π_1 by rotating Π_1 about the line l_1 by 90° . Find the vector equation of the plane Π_2 in scalar product form. [2]
- (d) The planes Π_3 and Π_4 have equations $x + 2y + 3z = 4$ and $\mathbf{r} \cdot \begin{pmatrix} 9 \\ 0 \\ 16 \end{pmatrix} = 21$ respectively. Find the vector equation of the line of intersection of the three planes Π_1 , Π_3 and Π_4 . [3]
- 2** The sequence of real numbers $a_1, a_2, a_3, \dots$ is an arithmetic progression.
- (a) Prove, by mathematical induction, that $\sum_{r=1}^n ra_r = \frac{n(n+1)(a_1 + 2a_n)}{6}$ for all $n \in \mathbb{Z}^+$. [4]
- (b) Hence, find an expression for
- (i) $\sum_{r=1}^n r^2$, in terms of n only. [2]
- (ii) $a_1 + 3a_2 + 5a_3 + 7a_4 + \dots + (2n-1)a_n$, in terms of n , a_1 and a_n only. [2]
- 3** The sequence of real numbers $b_1, b_2, b_3, \dots$ is such that $b_1 = 4$ and $b_n = b_{n-1} + 2n$ for all $n \in \mathbb{Z}^+$, $n \geq 2$. Using the method of difference, show that $b_n = n^2 + n + 2$ for all $n \in \mathbb{Z}^+$. [3]

- 4** The parametric equations of a curve are $x = t + \ln t$ and $y = t + e^t$, $t > 0$.
- (i) Sketch the curve, indicating clearly all intercepts and asymptotes. [3]
- (ii) Show that, for all the points on the curve, $\frac{dy}{dx} = \frac{t(1+e^t)}{t+1}$. Hence, deduce that the curve does not have any turning points. [3]
- (iii) Find, in exact form, the equation of the normal of the curve at the point where $t = 1$. [3]
- 5** (i) Sketch, on an Argand diagram, the set of points representing the complex number z which satisfy both conditions: $|z + 2i| \leq |z - 4 - 2i|$ and $0 \leq \arg(z + 2i) < \tan^{-1} 2$. [3]
- Hence, find the greatest value of $|z - 4i|$ which satisfy the above conditions, giving your answer in exact form. [1]
- (ii) With the help of your sketch in (i), sketch, on another Argand diagram, the set of points representing the complex number z which satisfy both conditions: $|z^* + 2i| \leq |z^* - 4 - 2i|$ and $0 \leq \arg(z^* + 2i) < \tan^{-1} 2$. [2]
- 6** (i) Let a be a positive real number. Find, in terms of a , the range of values of x for which the inequality $\frac{1}{x-a} > \frac{x}{x^2+1}$ is satisfied. [3]
- (ii) Hence, deduce the solution to the inequality $\frac{1}{|x|-a} > \frac{|x|}{x^2+1}$. [2]

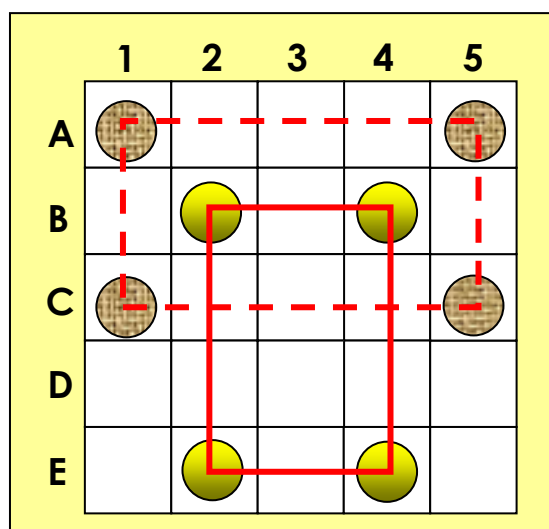
Section B : Statistics [60 marks]

- 7 Three students, Benn, Alvin and Reuben were tasked by the Entrepreneur Club to survey 50 students to collect data on the tee shirt sizes of students in the college so that the club can decide on how many tee shirts to stock up for different sizes. The school has 60 classes of 25 students each, of which 850 are female students. There are two levels with 800 students in year 1 and 700 students in year 2. The table shows the suggestions made by the three students on how they will collect the necessary data.

Student	Suggestion
Benn	I will conduct the survey during the lunch break from 11am to 2pm and survey every 30 th student entering the canteen about their tee shirt size.
Alvin	I will randomly select 50 classes and randomly select 1 student from each class for the survey.
Reuben	I will survey my own class as the class ratio of male to female student is similar to the school's ratio and I will survey another junior class with the similar male to female ratio.

Comment on the suitability of each of the student's approach for collection of the data. Suggest, with reasons, a more appropriate method (if any), to collect the necessary data. [5]

- 8 Yan, a toy creator, created a game to teach children the concept of geometrical shapes. The game consists of twenty-five cubicles arranged to form a big square box as shown. A child put four identical balls into four cubicles to form the four corners of a 'horizontal' rectangle (an example is indicated by the dotted lines in the diagram below) or a 'vertical' rectangle (an example is indicated by the solid lines in the diagram below). The labels at the sides of the boxes are used as a coordinate system.
- (i) Find the number of possible rectangles that can be formed if the four balls are placed in row A and row C. (Note that a square is a special rectangle.) [1]
- (ii) Yan claimed that 100 possible rectangles can be formed in this game. Prove that his claim is correct. [1]
- (iii) Find the number of rectangles that can be formed if one of the balls is in box B3. [2]
- (iv) Calculate the total possible number of squares that can be formed. [2]



- 9** The Entrepreneur Club is in charge of selling the school's tee shirt. Based on the sales record of the club, it was found that the monthly demand for tee shirt size XS has a Poisson distribution with mean 2 and the monthly demand for tee shirt size XXL has a Poisson distribution with mean 3. The club kept a monthly stock of 3 and 4 for tee shirt sizes XS and XXL respectively.
- (i) Calculate the probability that there is more than one XS size tee shirts being sold in a day assuming there are 30 days in a month. [2]
 - (ii) Calculate the probability that the club will not meet the demand for either XS or XXL tee shirts in a month. [2]
 - (iii) Find the most probable number of XXL tee shirts sold in a month. [2]
 - (iv) Determine the least number of stock needed each month for the XS tee shirts in order to meet the demand with a probability of at least 0.95. [2]
 - (v) The club earns a profit of \$5 for every tee shirt sold regardless of the size of the tee shirt. Assuming there is enough supply to meet all demand, use a suitable approximation to calculate the probability that the club will earn a profit of more than \$200 from the sales of the XS and XXL tee shirts in half a year. [3]
- 10** Kenny, a computer games addiction counsellor for schools, claimed that his single test for addiction to game is 99% accurate. From previous data, it is estimated that 2% of the student's population suffered from computer games addiction.
- (i) Show that the probability that a randomly chosen student is not a computer games addict and is diagnosed wrongly is 0.0098. [1]
 - (ii) Calculate the probability that a randomly chosen student will be diagnosed with addiction to computer games. [1]
 - (iii) An external counsellor doubted Kenny's method and commented that about 33.1% of those diagnosed with game addiction using Kenny's test actually do not have any addiction. Verify the validity of this statement. [2]
- To improve his method, Kenny will test those who were diagnosed as computer games addict a second time with 99% accuracy. Calculate the new probability that those diagnosed with game addiction actually do not have any addiction. [2]
- 11** The mathematics department of a college did an analysis on the preparation for lessons by students and discovered that 92% of the students were prepared for lessons.
- (i) Find the probability that in a group of 25 randomly chosen students, there were at least 20 students who were prepared for their lessons. [1]
 - (ii) A random sample of 50 students was selected. Using a suitable approximation to calculate the probability that at least 90% of the students were prepared for lessons. [3]

- 12** Two friends June and Cindy always meet in the library to study together. Their arrival time to the library and length of stay in the library is independent of each other. June's arrival time follows a normal distribution with mean arrival time at 1700h with a standard deviation of 30 minutes and her stay in the library follows a normal distribution with mean 40 minutes and standard deviation 15 minutes. On the other hand, Cindy will reach the library every day at 1730h sharp and her stay in the library follows a normal distribution with mean 1 hour and standard deviation 15 minutes.
- (i) On a randomly chosen day, calculate the following probabilities:
- (a) Cindy is in the library earlier than June. [2]
(b) June leaves before Cindy arrives. [2]
- (ii) Calculate the probability that the total time spent by June for three days in the library is more than three times the time spent by Cindy in a day. [2]

For a particular week, Cindy is not consistent in turning up at the library. The probability that Cindy will visit the library on any particular day is 0.7. If she visits the library, she will turn up at 1730h sharp and her stay will follow the same distribution as before.

- (iii) Show that the probability that June and Cindy will not meet in a particular day is 0.5705. [3]

- 13** The education department of a district analysed the performance of the students in their district for an international examination. The mean marks for the international examination is 84. The marks, x , of 500 randomly chosen students were tabulated and the results are summarised as follows :

$$\sum x = 41485 \quad , \quad \sum x^2 = 3524283 .$$

- (i) Calculate the unbiased estimates for the population mean and the population variance of the marks for the students in the district, giving your answer to 2 decimal places. [2]
- (ii) Find the maximum level of significance such that we do not reject the claim that the students' performance had not deteriorated. [3]
- (iii) Given that n random samples of 500 students were collected. Calculate the minimum n such that at 5% level of significance, the probability of having at least one case of rejecting null hypothesis wrongly is at least 0.99. [3]
- (iv) At a later date, it was revealed that the standard deviation for the marks in the international examination is 12. Calculate the probability that the mean marks for a random sample of 50 students differs from the mean mark of 84 by more than 4 marks. [3]

- 14** The data below showed the average monthly Brent Oil Price and the average monthly exchange rate for each US dollar to Singapore Dollar for ten months from August 2007 to May 2008.

Time (t months)	1	2	3	4	5	6	7	8	9	10
Brent Oil Price (x \$US per Barrel)	70.76	77.17	82.34	92.41	90.93	92.18	94.99	103.64	109.07	122.80
Exchange rate 1\$US to Singapore Dollar (y)	1.52	1.51	1.46	1.45	1.45	1.43	1.41	1.38	1.36	1.37
<i>Source:</i> -Energy Information Administration, US -Federal Reserve Statistical Release of the United States										

It is given that the value of the product moment correlation coefficient between t and y is -0.977 and the equation of regression line of y on t is $y = 1.532 - 0.0178t$.

- (i) Find the least square regression line of x on t . Sketch this line on the scatter diagram of x against t . [2]
- (ii) Calculate the product moment correlation coefficient between t and x . [1]
- (iii) Use a regression line to estimate the time when $x = 130$ and comment on the result. [3]
- (iv) Comment on the implications for Singapore given the present trend of the exchange rates and the oil prices. [2]

~ End of Paper ~